

ANALYSTLAB AFRICA
EXPERIENCE LAB INTERNSHIP PROGRAMME
HEALTHCONNECT CLINIC EXPERIENCE LAB
WEEK 6 — UPDATED RISK & DEPENDENCY REGISTER

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Field	Detail
Project	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Programme	AnalystLab Africa Experience Lab
Control Function	Project Management
Reporting Period	Week 6
Version	2.0
Status	Controlled — Active
Next Review Point	Week 7 Testing & Refinement

AMBER — CONTROLLED RISK & DEPENDENCY EXPOSURE

1. Register Purpose

This register provides the consolidated control position for project risks and material delivery dependencies at the transition from Week 6 Integration, Advanced Development and Validation into Week 7 Testing & Refinement.

It is intended to answer four control questions:

1. What could still materially affect delivery or readiness?
2. What conditions must hold for downstream work to proceed?
3. What controls are currently reducing the exposure?
4. What evidence is required before the exposure can be reduced, accepted or closed?

The register is deliberately evidence-led. Completion of an activity does not, by itself, reduce a risk or satisfy a dependency.

A risk remains open where the underlying exposure persists. A dependency remains open where the required downstream condition has not yet been demonstrably satisfied.

2. Control Boundary

This register distinguishes four related but different control conditions:

Control Type	Meaning
Risk	An uncertain condition that may affect project objectives if it materialises

Control Type	Meaning
Dependency	A required relationship, input, handoff or condition on which downstream progress depends
Issue	A condition that has already occurred and requires active resolution
Decision	A controlled determination that establishes direction, treatment or acceptance

The register therefore does not replace the:

- Integration Issue Log;
- Decision Log;
- Cross-Track Coordination Record;
- Project Status Update;
- Week 7 Testing & Refinement Coordination Plan; or
- detailed Integration & Dependency Register.

Where another control document contains the authoritative event or record, this register captures only the risk exposure or dependency consequence relevant to project control.

3. Risk Assessment Standard

3.1 Probability

Rating	Definition
1 — Rare	Unlikely under current conditions
2 — Unlikely	Possible, but not expected
3 — Possible	Credible under current conditions
4 — Likely	More likely than not without further control
5 — Almost Certain	Expected without effective intervention

3.2 Impact

Impact considers the highest credible consequence across: Schedule, Quality/Evidence, Integration, Readiness, and Governance/Safety.

Rating	Definition
1 — Minor	Localised effect with negligible project consequence
2 — Low	Manageable effect on an activity or isolated output
3 — Moderate	Material effect requiring coordinated management
4 — Major	Significant downstream or cross-workstream consequence
5 — Severe	Threat to a material project objective, safety condition or progression gate

3.3 Risk Rating

$$\text{Risk Score} = \text{Probability} \times \text{Impact}$$

Score	Rating	Required Control
1–4	Low	Monitor through normal control cycle
5–9	Moderate	Active mitigation and periodic review
10–16	High	Defined mitigation, owner accountability and formal PM review
17–25	Critical	Immediate escalation and management intervention

This scoring model is applied consistently to avoid numerical ambiguity. On the current assessment, no risk is formally rated Critical.

4. Risk Register — Current Control Position

ID	Risk Exposure	Principal Cause / Trigger	Potential Project Consequence	P	I	Score	Rating	Treatment / Control	Residual Position	Owner	Status
R-01	Data limitations may restrict the generalisability of analytical and predictive conclusions	Limited subgroup sizes, synthetic/fixed-period data and absence of future operational data	Findings or model behaviour may not transfer reliably to future populations	3	4	12	HIGH	Maintain dataset limitations in downstream evidence; distinguish observed patterns from generalisable claims; require limitations in testing and acceptance records	Moderate exposure remains	Data Analytics / Data Science	Monitoring
R-02	Analytical associations may be interpreted as causal or predictive evidence	Multiple subgroup findings and statistical significance may be overstated	Incorrect intervention assumptions or inappropriate modelling priorities	3	4	12	HIGH	Preserve association/causality distinction; require claims to remain traceable to supporting evidence; use predictive evidence before ranking variables as model drivers	Moderate exposure remains	Data Analytics / PM	Active
R-03	Predictive development may fail to establish a defensible final candidate	Week 6 comparative model execution and evaluation remain outstanding	Candidate selection and subsequent technical integration may be delayed or unsupported	4	4	16	HIGH	Complete comparative evaluation against defined metrics; document candidate-selection rationale, limitations and error behaviour before handoff	High until comparative evidence is produced	Data Science	Active
R-04	The selected DS candidate may not be transferred to ML within the testing sequence	Final candidate selection and formal DS→ML handoff remain outstanding	Actual cross-workstream integration remains unverified; testing may rely on stand-in evidence	4	4	16	HIGH	Prioritise candidate selection, handoff package and receiver confirmation; retain RF stand-in evidence only as technical pathway evidence	High until actual candidate is integrated	Data Science / ML	Active
R-05	ML scoring capability may remain technically functional without an agreed operational interpretation	Threshold/action interpretation has not yet been established	Risk scores may be technically available but unsuitable for operational decision-making	3	4	12	HIGH	Maintain explicit separation between technical scoring and operational readiness; define intended interpretation and acceptance conditions	Moderate–High	ML / PM	Active
R-06	Analytics→DS downstream utilisation may remain insufficiently evidenced	Findings/features were exchanged, but resulting downstream incorporation is not fully traceable	Analytical work may not demonstrably influence predictive development or model decisions	3	4	12	HIGH	Require evidence of received input, incorporated change and resulting effect before classifying interface as fully integrated	High until downstream evidence is established	Data Analytics / Data Science	Active
R-07	Predictive target definition may remain ambiguous regarding Cancelled appointments	Binary Attended/No-Show direction adopted while treatment of Cancelled remains unresolved	Inconsistent training population, interpretation or acceptance criteria	3	4	12	HIGH	Resolve treatment before final candidate acceptance; record rationale and downstream implications	Moderate–High	Data Science / PM	Active
R-08	GenAI safety-boundary behaviour may remain inconsistent across conversational contexts	Emergency-disclaimer inconsistency identified during targeted testing; broader scenario coverage remains limited	Insufficient confidence in behavioural reliability for wider testing or operational use	3	5	15	HIGH	Targeted refinement, regression testing and broader scenario coverage; retain the limitation until retest evidence demonstrates improvement	High until verified	Generative AI	Active

ID	Risk Exposure	Principal Cause / Trigger	Potential Project Consequence	P	I	Score	Rating	Treatment / Control	Residual Position	Owner	Status
R-09	GenAI service interruptions may reduce testing coverage	Intermittent 503/API responses disrupted portions of testing	Scenario coverage or repeatability of evidence may be reduced	3	3	9	MODERATE	Record interrupted tests; repeat material scenarios where necessary; distinguish infrastructure interruption from behavioural result	Low–Moderate if service stabilises	Generative AI / PM	Monitoring
R-10	Workstreams may enter testing at materially different levels of readiness	Analytics is comparatively mature while DS, ML and GenAI retain conditional readiness conditions	Testing sequence may require resequencing, repeated coordination or conditional acceptance	4	4	16	HIGH	Use dependency-driven sequencing; permit independent testing where appropriate while protecting critical interfaces	Moderate–High	PM / All Workstreams	Active
R-11	Proposed interventions may not adequately reflect the behavioural evidence	Analytical relationships are observational and intervention effectiveness has not yet been demonstrated	Resources may be directed toward interventions whose effectiveness remains uncertain	3	4	12	HIGH	Treat intervention proposals as evidence-based hypotheses until tested; require validation before effectiveness claims	Moderate	PM / Data Analytics / Data Science	Monitoring
R-12	Late-emerging acceptance requirements may generate avoidable rework	Deeper Week 7 testing may expose evidence, interface or acceptance gaps	Additional refinement, retesting or delayed readiness determination	3	3	9	MODERATE	Establish acceptance dimensions before testing; route newly identified conditions through controlled issue/decision processes	Low–Moderate	PM / All Workstreams	Active
R-13	Differences in interpretation or readiness thresholds may delay cross-workstream decisions	Workstreams assess technical completion using different criteria	Conflicting readiness positions or delayed acceptance decisions	2	4	8	MODERATE	Maintain common acceptance framework; escalate only where disagreement affects material delivery or readiness	Low–Moderate	PM / Workstream Leads	Monitoring
R-14	Concentration of outstanding interfaces may increase coordination pressure during Week 7	Several material handoffs and refinement activities converge within the testing window	Reduced turnaround time and increased sequencing pressure	3	3	9	MODERATE	Prioritise material interfaces, establish control checkpoints and escalate only where coordination produces material consequence	Low–Moderate	PM	Monitoring

5. Material Risk Interpretation

The register indicates that the project's principal exposure is no longer broad project uncertainty. It is concentrated around evidence maturity and interface completion.

Highest current exposures

Predictive pathway:

- Comparative DS evaluation;
- Candidate selection;
- DS→ML transfer;
- Actual candidate integration.

Evidence pathway:

- Analytics→DS downstream traceability;
- Distinction between statistical association and predictive/causal claims;
- Treatment of unresolved target-definition questions.

Operational pathway:

- Interpretation of ML risk scores;
- Distinction between technical capability and operational readiness.

GenAI pathway:

- Emergency-disclaimer refinement;
- Regression evidence;
- Broader behavioural coverage.

These exposures are interrelated but should not be collapsed into one generic “integration risk.” Their control requirements and evidence thresholds differ.

6. Risk Response Register

Risk	Response Strategy	Immediate Control Objective	Closure / Reduction Evidence
R-01	Reduce / Monitor	Keep data limitations visible in downstream interpretation	Limitations documented and accepted within testing/readiness evidence
R-02	Reduce	Prevent unsupported causal or predictive claims	Revised interpretation supported by appropriate evidence
R-03	Reduce	Establish defensible candidate through executed comparison	Comparative results + selection rationale + limitations
R-04	Reduce / Escalate if material	Complete actual DS→ML transfer	Candidate package received, accepted and integrated

Risk	Response Strategy	Immediate Control Objective	Closure / Reduction Evidence
R-05	Reduce	Define meaning and intended use of ML score	Operational interpretation and acceptance condition documented
R-06	Reduce	Establish downstream analytical incorporation	Receiver-side evidence showing resulting change
R-07	Reduce	Close target-definition ambiguity	Documented treatment of Cancelled records and implications
R-08	Reduce / Retest	Establish consistent safety-boundary behaviour	Corrected behaviour demonstrated through repeat testing
R-09	Monitor / Reduce	Protect testing evidence from infrastructure interruptions	Required scenarios successfully repeated
R-10	Reduce	Align readiness conditions with testing sequence	Material interfaces sufficiently ready for planned testing
R-11	Reduce	Prevent intervention assumptions from being treated as validated effects	Intervention evidence or explicit hypothesis status
R-12	Reduce	Minimise late acceptance rework	Acceptance criteria applied before final testing
R-13	Reduce	Maintain common readiness interpretation	Material disagreements resolved or formally dispositioned
R-14	Reduce	Protect PM coordination capacity around critical interfaces	Material coordination actions completed within testing sequence

7. Risk Triggers

A risk requires reassessment when any of the following occurs:

- A mitigation fails or becomes ineffective;
- The probability or consequence materially changes;
- A dependency becomes delayed, unavailable or rejected;
- An expected output is not received;
- An acceptance condition becomes unattainable;
- A risk materialises into an actual issue;
- Repeated testing reveals the same unresolved condition;
- A technical change creates a new downstream exposure;
- A safety or governance concern emerges; or
- The risk threatens a Week 7 control gate.

A materialised risk shall transition into the Issue Log rather than remaining artificially classified as a future risk.

8. Dependency Control Framework

Dependencies are assessed using the following maturity states:

State	Definition
Identified	Relationship recognised but output not yet produced
Produced	Required output exists
Transferred	Output has been passed to the receiving workstream
Accepted for Use	Receiver confirms the output is sufficiently usable
Integrated	Output has demonstrably changed or contributed to downstream work
Validated	Integrated output has passed the relevant validation condition
Testing Ready	Output satisfies conditions required to enter planned testing
Closed	Dependency no longer requires active control

Critical distinction: production, transfer, acceptance, integration and validation are separate states.

9. Material Dependency Register — Week 6 Position

ID	Provider	Receiver	Dependency / Required Condition	Current State	Required Evidence	Risk Link	Week 7 Position
D-01	PM	All Workstreams	Approved baseline, scope and requirements	Controlled	Current baseline traceability	R-10	Maintained
D-02	PM / Requirements	All Workstreams	Validated problem definition and requirements	Controlled	Problem/requirement baseline	R-10	Maintained
D-03	Data Analytics	Data Analytics	Available and quality-assessed appointment data	Produced / Controlled	Data-quality evidence	R-01	Available
D-04	Data Analytics	Data Analytics	Analysis-ready dataset	Produced / Accepted	Cleaned and enriched dataset	R-01	Available
D-05	Data Analytics	Data Science	Validated attendance/no-show findings	Transferred; downstream incorporation not fully evidenced	Receiver-side use and resulting change	R-06	Requires confirmation
D-06	Data Analytics	ML	Relevant analytical/model-input requirements where required	Conditional	Actual downstream use	R-10	Not a primary active handoff
D-07	Data Science	ML	Selected candidate model, specifications and implementation requirements	Awaiting final candidate	Candidate package + receiver acceptance	R-03 / R-04	Priority dependency
D-08	ML	Data Science	Technical feedback / integrated implementation evidence	Conditional	Verified technical feedback against actual candidate	R-04	Dependent on D-07
D-09	Data Analytics	Generative AI	Evidence-based support insights where support design requires them	Conditional / Not Activated	Demonstrable downstream requirement and use	R-11	Not established as active technical integration
D-10	Generative AI	Integrated Solution	Support capability, knowledge boundaries and escalation requirements	Produced / Conditionally Ready	Behavioural evidence + acceptance conditions	R-08	Requires refinement evidence
D-11	All Workstreams	PM	Outputs, status and dependency evidence	Active	Traceable workstream evidence	R-10 / R-14	Ongoing
D-12	All Workstreams	Integrated Delivery	Confirmed material cross-track handoffs	Partial	Evidence of transfer, acceptance and downstream use	R-10	Open
D-13	Data Science	ML	Updated candidate model and implementation package	Pending	Selected candidate + transfer package	R-03 / R-04	Priority dependency
D-14	ML	Integrated Solution	Technical component/interface evidence	Partially evidenced	Actual integrated component + verification	R-05 / R-10	Conditional

ID	Provider	Receiver	Dependency / Required Condition	Current State	Required Evidence	Risk Link	Week 7 Position
D-15	Generative AI	Integrated Solution	Refined assistant and safety/testing evidence	Conditional	Retest and regression evidence	R-08	Priority refinement dependency
D-16	All Workstreams	Week 7 Testing	Testable outputs, known limitations and acceptance conditions	Future / Planned	Test evidence and acceptance criteria	R-10 / R-12	Entry dependency

10. Dependency Exceptions

Three dependency conditions require explicit control rather than being represented as routine handoffs.

D-05 — Analytics → Data Science

The exchange of validated findings/features has been reported. However, the available evidence does not yet establish the complete chain:

Input received → incorporated into downstream work → resulting change → accepted for use.

Accordingly, this dependency should not yet be classified as fully integrated.

D-07 / D-13 — Data Science → ML

This remains the most consequential technical handoff. ML has demonstrated that its scoring and interface pathway can function using a Random Forest stand-in. That evidence establishes technical pathway readiness, not actual DS→ML integration.

The dependency remains open until the selected DS candidate is transferred and verified through the ML interface.

D-15 — Generative AI → Integrated Solution

GenAI has produced refinement and testing evidence, but the evidence currently supports conditional testing readiness, not full integrated-solution readiness.

The emergency-disclaimer inconsistency and limited breadth of behavioural evidence remain relevant conditions for Week 7.

11. Dependency Acceptance Standard

A dependency shall not be considered complete solely because the provider reports that the output has been produced.

Where material, acceptance should demonstrate:

5. Output existence — the required artefact or capability exists.
6. Transfer — the intended receiver has received it.
7. Usability — the receiver can use it for the intended downstream purpose.
8. Incorporation — downstream use or integration is demonstrable.
9. Resulting change — the effect of the handoff can be identified.
10. Evidence — the claim is supported by an appropriate record.
11. Readiness effect — the dependency's impact on the next stage is understood.

This prevents “handoff completed” from being treated as equivalent to “integration completed.”

12. Risk and Dependency Interface

The relationship between the two registers is controlled as follows:



For example:



This structure prevents the same event from being unnecessarily duplicated across control records.

13. Escalation Control

Escalation is required where an unresolved risk or dependency:

- Blocks a downstream workstream;
- Threatens a Week 7 control gate;
- Materially affects evidence quality;
- Creates a safety or governance concern;
- Prevents acceptance of a material output;
- Repeatedly fails despite mitigation;
- Exceeds workstream authority; or
- Creates a material schedule, integration or readiness consequence.

Escalation is therefore based on project consequence, not simply technical complexity.

Escalation Levels

Level	Meaning	PM Action
Level 1 — Monitor	Exposure controlled within workstream	Maintain visibility
Level 2 — Manage	Cross-workstream coordination required	Coordinate mitigation and evidence
Level 3 — Escalate	Material consequence or authority limit	Escalate through project governance

14. Risk Ageing and Closure Control

An open risk shall remain visible until there is sufficient evidence that:

- The exposure has materially reduced;
- The condition is no longer relevant;
- The exposure has been formally accepted; or
- The risk has materialised and transferred to issue management.

Risk closure shall therefore require disposition evidence, not task completion.

Similarly, a dependency shall not be closed merely because the provider has delivered an output. Where downstream use remains outstanding, the dependency remains active.

15. Week 6 Change Assessment

Compared with the previous risk/dependency baseline, the Week 6 control position has changed in the following respects:

Area	Previous Exposure	Week 6 Position
Analytical evidence	General analytical uncertainty	Evidence strengthened through statistical validation and deeper interaction analysis
Predictive development	General model-performance risk	More specific exposure around comparative execution and candidate selection
DS→ML	Future technical dependency	Material active handoff risk; stand-in pathway tested, actual candidate not integrated
ML	Training/scoring capability risk	Technical scoring pathway strengthened; operational interpretation remains unresolved
Analytics→DS	General analytical dependency	Exchange reported; downstream incorporation remains insufficiently traceable
GenAI	Boundary and escalation uncertainty	Targeted testing produced concrete behavioural evidence and identified disclaimer inconsistency
Cross-workstream integration	General alignment risk	Exposure now concentrated around actual interface completion and evidence transfer
Week 7 readiness	Future testing uncertainty	Controlled testing can proceed, subject to defined entry conditions

This is the principal purpose of the Week 6 update: to replace broad anticipated risks with risks anchored to observed integration evidence and remaining conditions.

16. Week 7 Control Priorities

The risk and dependency register identifies the following sequence as the highest-control pathway:



Independent testing may proceed in parallel where the dependency structure permits it. However, parallel activity must not be used to imply that an unresolved downstream dependency has been satisfied.

17. Current Risk & Dependency Position

OVERALL CONTROL POSITION: AMBER — CONTROLLED WITH MATERIAL CONDITIONS

The project remains capable of progressing into Week 7 testing and refinement. However, the evidence does not yet support an unrestricted readiness position.

The principal conditions are

- Actual DS→ML candidate transfer;
- Evidence-backed predictive candidate selection;
- Analytics→DS downstream traceability;
- Resolution of predictive target-definition ambiguity;
- GenAI disclaimer refinement and retest;
- Operational interpretation of ML risk scores; and
- Continued separation of technical, integrated, validated and operational readiness.

The current exposure is therefore controlled rather than eliminated.

18. PM Control Determination

From a project-management perspective, the Week 6 evidence establishes meaningful progress in risk reduction and dependency clarification.

It does not establish that all material interfaces are integrated or that all downstream outputs are ready for operational acceptance.

Proceed to Week 7 Testing & Refinement under controlled conditions. Maintain heightened monitoring of the DS→ML handoff, predictive candidate evidence, Analytics→DS traceability, GenAI behavioural refinement and ML operational interpretation.

The project should not convert technical completion, communication, or simulated integration evidence into claims of full solution readiness.

19. Governance and Maintenance

The PM function shall maintain this register as a living control record. At each formal review, PM shall reassess:

- Probability;
- Impact;
- Risk rating;
- Treatment effectiveness;
- Residual exposure;
- Dependency state;
- Material changes;
- Escalation requirement;
- Issue conversion;
- Closure evidence; and
- Implications for project readiness.

Workstream owners remain accountable for implementing agreed mitigation within their technical remit.

PM remains accountable for visibility, control integrity, cross-workstream coordination, escalation and project-level risk disposition.

20. Control Document Interface

Document	Control Purpose	Relationship to This Register
Integration & Dependency Register	Detailed dependency relationships and conditions	Source for dependency relationships
Integration Issue Log	Material conditions already occurring	Receives risks that materialise into issues
Decision Log	Controlled project decisions	Records formal decisions arising from risk/dependency exposure
Cross-Track Coordination Record	Actual exchanges, handoffs and resulting integration activity	Provides evidence for dependency status
Project Status Update	Overall project health and management position	Receives consolidated risk position
Week 7 Testing & Refinement Plan	Testing, refinement and acceptance pathway	Defines conditions against which several risks will be reassessed

This separation preserves single-purpose governance records and prevents duplication.

21. Version Control

Version	Period	Status	Change
1.0	Week 5	Baseline	Initial risk and dependency framework established
2.0	Week 6	Current	Risk exposure reassessed against actual integration evidence; material dependencies reclassified; residual exposure, acceptance conditions and Week 7 controls established

Document Owner: Project Management. Current Status: AMBER — Controlled Risk & Dependency Exposure. Next Review: Week 7 Testing & Refinement.